

ISC Class XII Physics

Chapter 10 Wave Optics

36 Numericals Answer Key

Step-by-Step ISC Board Solutions

Topics Covered

- I. Reflection and Refraction of Light Waves
- II. Superposition of Waves
- III. Young's Double Slit Experiment (YDSE)
- IV. Angular Fringe Width
- V. Diffraction at a Single Slit
- VI. Miscellaneous (Combined YDSE / Diffraction Problems)

Contents

1	I. Reflection and Refraction of Light Waves	7
2	II. Superposition of Waves	13
3	III. Young's Double Slit Experiment (YDSE)	15
4	IV. Angular Fringe Width	21
5	V. Diffraction at a Single Slit	22
6	VI. Miscellaneous	25

Index of Numericals

Q. No.	Topic / Statement Summary	Section
1	Speed of light in a medium of refractive index 1.5	I
2	Time taken by light to cross 1.0 m thickness of glass	I
3	Difference in speed of light in glass for violet and red light	I
4	Speed of light in glass, given speed of light in water	I
5	Wavelength of light inside a glass block	I
6	Frequency and speed of light in water	I
7	Refractive index of B w.r.t. A from equal transit time	I
8	Frequency, period and wavelength in glass for mercury green line	I
9	Wavelength in vacuum, refractive index and speed in a liquid	I
10	Wavelength, frequency of reflected light; angle for perpendicular reflection	I
11	Speed, wavelength, frequency of reflected and refracted yellow light	I
12	Ratio of maximum to minimum intensity of two superposing waves	II
13	Ratio of intensities of maxima and minima from coherent sources	II

Q. No.	Topic / Statement Summary	Section
14	Ratio of resultant intensities at two points with given path differences	II
15	Total phase difference from initial phase and path difference	III
16	Screen distance for a given fringe width; new fringe width if doubled	III
17	Fringe width; effect of halving d and doubling D	III
18	Phase and path difference at 3rd bright and 3rd dark fringe	III
19	Wavelength and fringe width from position of 4th dark fringe	III
20	Slit separation from frequency and fringe width	III
21	Wavelength of light from fringe-width data	III
22	Distance between first dark fringe and first bright fringe	III
23	Path difference at the position of first-order maximum	III
24	Angular position of 10th bright fringe; angular fringe width	IV
25	Angular width of first minimum and first subsidiary maxima	V
26	Wavelength from position of first minima (Fraunhofer pattern)	V
27	Total angular and linear width of central diffraction maximum	V
28	Second wavelength from coinciding minimum/maximum condition	V
29	New fringe spacing after changing D , d and λ together	VI
30	Wavelength from width of 10 fringes; new width at larger D	VI
31	Wavelength of second source by comparing fringe counts	VI
32	Distance between fifth bright fringe and first dark fringe	VI
33	Wavelength from dark-fringe position; effect of immersion in water	VI
34	Width of central maximum in air and immersed in water	VI
35	Slit width for ten double-slit maxima within central maximum	VI
36	Width of central maximum formed using a convex lens	VI

Standard values used throughout this chapter: Speed of light in vacuum/air, $c = 3.0 \times 10^8 \text{ m s}^{-1}$; $1 \text{ \AA} = 10^{-10} \text{ m}$.

Formula Sheet

I. Reflection and Refraction of Light Waves

Speed in a medium

$$v = \frac{c}{n}$$

c = speed in vacuum, n = refractive index of medium, v = speed in medium

Time to traverse a thickness

$$t = \frac{d}{v}$$

d = thickness of medium, v = speed of light in that medium

Wavelength in a medium

$$\lambda_{\text{medium}} = \frac{\lambda_{\text{vacuum}}}{n}$$

Frequency does not change on entering a new medium; only v and λ change

Relative refractive index

$${}_A n_B = \frac{v_A}{v_B} = \frac{n_B}{n_A}$$

Refractive index of medium B with respect to medium A

Frequency–wavelength relation

$$f = \frac{c}{\lambda}$$

f = frequency, λ = wavelength in vacuum/air

II. Superposition of Waves

Intensity–amplitude relation

$$I \propto a^2$$

I = intensity, a = amplitude of the wave

Max. to min. intensity ratio

$$\frac{I_{max}}{I_{min}} = \left(\frac{a_1 + a_2}{a_1 - a_2} \right)^2$$

a_1, a_2 = amplitudes of the two superposing waves

Resultant intensity (phase form)

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

ϕ = phase difference between the two waves at the point of interest

III. Young's Double Slit Experiment (YDSE)**Fringe width**

$$\beta = \frac{\lambda D}{d}$$

D = slit-to-screen distance, d = slit separation, λ = wavelength

Position of n th bright fringe

$$x_n = \frac{n\lambda D}{d}$$

$n = 0, 1, 2, \dots$ measured from the central bright fringe

Position of n th dark fringe

$$x_n = \frac{(2n - 1)\lambda D}{2d}$$

$n = 1, 2, 3, \dots$ measured from the central bright fringe

Path difference

$$\Delta x = d \sin \theta \approx \frac{xd}{D}$$

θ = angle subtended at the slits; valid for small θ

Phase difference

$$\phi = \frac{2\pi}{\lambda} \Delta x$$

Relates path difference Δx to phase difference ϕ

IV. Angular Fringe Width

Angular position of n th bright fringe

$$\theta_n = \frac{n\lambda}{d}$$

Small-angle form of x_n/D

Angular fringe width

$$\Delta\theta = \frac{\lambda}{d}$$

Angular separation between two consecutive bright (or dark) fringes

V. Diffraction at a Single Slit

Minima condition

$$a \sin \theta = n\lambda$$

a = slit width, $n = 1, 2, 3, \dots$

Secondary (subsidiary) maxima

$$a \sin \theta = (2n + 1) \frac{\lambda}{2}$$

Approximate condition, $n = 1, 2, 3, \dots$

Angular width of central maximum

$$2\theta = \frac{2\lambda}{a}$$

Angle between the first minima on either side of the central maximum

Linear width of central maximum

$$w = \frac{2\lambda D}{a}$$

D = slit-to-screen distance (or focal length f of the focusing lens)

VI. Miscellaneous (Combined Relations)

Double-slit maxima inside single-slit envelope

$$N = \frac{2d}{e}$$

N = number of double-slit maxima within the central diffraction maximum, e = slit width,
 d = slit separation

Central maximum width using a lens

$$w = \frac{2f\lambda}{a}$$

Screen placed at focal plane of a convex lens of focal length f kept close to the slit

I. Reflection and Refraction of Light Waves

Question 1

What will be the speed of light in a medium of refractive index 1.5?

★ Likely Exam Question

Given: $n = 1.5$, $c = 3.0 \times 10^8 \text{ m s}^{-1}$

Formula: $v = \frac{c}{n}$

Solution:

$$v = \frac{3.0 \times 10^8}{1.5} = 2.0 \times 10^8 \text{ m s}^{-1}$$

Final Answer

$$v = 2.0 \times 10^8 \text{ m s}^{-1}$$

Common Mistake: Do not confuse refractive index with speed itself – n is dimensionless; always divide c by n , never multiply.

Question 2

Find the time taken by light to travel 1.0 m thickness of glass whose refractive index is 1.5.

Given: $d = 1.0 \text{ m}$, $n = 1.5$, $c = 3.0 \times 10^8 \text{ m s}^{-1}$

Formula: $v = \frac{c}{n}$, $t = \frac{d}{v}$

Solution:

$$v = \frac{3.0 \times 10^8}{1.5} = 2.0 \times 10^8 \text{ m s}^{-1}$$
$$t = \frac{d}{v} = \frac{1.0}{2.0 \times 10^8} = 5.0 \times 10^{-9} \text{ s} = 5 \text{ ns}$$

Final Answer

$$t = 5 \text{ ns}$$

Question 3

The refractive index of glass for violet light is 1.665 and for red light is 1.618. Compute the difference in the speed of light in glass for the two colours.

Given: $n_v = 1.665$, $n_r = 1.618$, $c = 3.0 \times 10^8 \text{ m s}^{-1}$

Formula: $v = \frac{c}{n}$

Solution:

$$v_v = \frac{3.0 \times 10^8}{1.665} = 1.8018 \times 10^8 \text{ m s}^{-1}$$

$$v_r = \frac{3.0 \times 10^8}{1.618} = 1.8541 \times 10^8 \text{ m s}^{-1}$$

$$\Delta v = v_r - v_v = 0.0523 \times 10^8 = 5.2 \times 10^6 \text{ m s}^{-1}$$

Final Answer

$$\Delta v \approx 5.2 \times 10^6 \text{ m s}^{-1}$$

Common Mistake: Since $n_v > n_r$, violet light travels *slower* in glass than red light – always subtract the smaller speed from the larger one.

Question 4

The refractive index of glass is 1.5 and that of water is 1.3. If the speed of light in water is $2.25 \times 10^8 \text{ m s}^{-1}$, find the speed of light in glass.

Given: $n_g = 1.5$, $n_w = 1.3$, $v_w = 2.25 \times 10^8 \text{ m s}^{-1}$

Formula: $c = n_w v_w$, $v_g = \frac{c}{n_g}$

Solution:

$$c = n_w v_w = 1.3 \times 2.25 \times 10^8 = 2.925 \times 10^8 \text{ m s}^{-1}$$

$$v_g = \frac{c}{n_g} = \frac{2.925 \times 10^8}{1.5} = 1.95 \times 10^8 \text{ m s}^{-1}$$

Final Answer

$$v_g = 1.95 \times 10^8 \text{ m s}^{-1}$$

Common Mistake: Find c first from the water data – do not assume the standard $3.0 \times 10^8 \text{ m s}^{-1}$ value when the question itself gives you enough information to calculate it.

Question 5

Light of wavelength $4500 \text{ \AA}$ in vacuum enters into a glass block of refractive index 1.5. What is the wavelength of light in the glass block?

Given: $\lambda_{\text{vac}} = 4500 \text{ \AA}$, $n = 1.5$

Formula: $\lambda_{\text{medium}} = \frac{\lambda_{\text{vacuum}}}{n}$

Solution:

$$\lambda_{\text{glass}} = \frac{4500}{1.5} = 3000 \text{ \AA}$$

Final Answer

$$\lambda_{\text{glass}} = 3000 \text{ \AA}$$

Question 6

The wavelength of monochromatic light in air is $6000 \text{ \AA}$. The wavelength of this light in water is $4500 \text{ \AA}$. Find (i) frequency of light in water, (ii) speed of light in water.

★ Likely Exam Question

Given: $\lambda_{\text{air}} = 6000 \text{ \AA}$, $\lambda_{\text{water}} = 4500 \text{ \AA}$, $c = 3.0 \times 10^8 \text{ m s}^{-1}$

Formula: $f = \frac{c}{\lambda_{\text{air}}}$ (frequency unchanged across media), $v = f\lambda_{\text{water}}$

Solution:

$$f = \frac{3.0 \times 10^8}{6000 \times 10^{-10}} = 5.0 \times 10^{14} \text{ Hz}$$

$$v_{\text{water}} = f \times \lambda_{\text{water}} = 5.0 \times 10^{14} \times 4500 \times 10^{-10} = 2.25 \times 10^8 \text{ m s}^{-1}$$

Final Answer

(i) $f = 5.0 \times 10^{14} \text{ Hz}$ (same as in air) (ii) $v = 2.25 \times 10^8 \text{ m s}^{-1}$

Common Mistake: Frequency never changes when light passes from one medium to another – only wavelength and speed change. Students often wrongly recompute frequency using λ_{water} and the vacuum value of c .

Question 7

The ratio of thicknesses of the strips of two transparent media A and B is $6 : 4$. If light takes the same time in passing through both of them, find the refractive index of B with respect to A .

Given: $d_A : d_B = 6 : 4 = 3 : 2$, $t_A = t_B$

Formula: $t = \frac{d}{v}$, ${}^A n_B = \frac{v_A}{v_B}$

Solution:

Since the time is the same,

$$\frac{d_A}{v_A} = \frac{d_B}{v_B} \implies \frac{v_A}{v_B} = \frac{d_A}{d_B} = \frac{3}{2}$$

$${}^A n_B = \frac{v_A}{v_B} = 1.5$$

Final Answer

Refractive index of B with respect to $A = 1.5$

Common Mistake: Refractive index is inversely proportional to speed, so make sure you take the ratio of speeds (not thicknesses directly) as the final answer.

Question 8

Green light of mercury has a wavelength of 5.5×10^{-7} m in air. (i) What is its frequency in MHz and period in μ s? (ii) What is its wavelength in glass if $n_{\text{glass}} = 1.5$?

Given: $\lambda_{\text{air}} = 5.5 \times 10^{-7}$ m, $n_{\text{glass}} = 1.5$, $c = 3.0 \times 10^8$ m s $^{-1}$

Formula: $f = \frac{c}{\lambda}$, $T = \frac{1}{f}$, $\lambda_{\text{glass}} = \frac{\lambda_{\text{air}}}{n}$

Solution:

$$f = \frac{3.0 \times 10^8}{5.5 \times 10^{-7}} = 5.45 \times 10^{14} \text{ Hz} = 5.45 \times 10^8 \text{ MHz}$$

$$T = \frac{1}{f} = \frac{1}{5.45 \times 10^{14}} = 1.83 \times 10^{-15} \text{ s} = 1.83 \times 10^{-9} \mu\text{s}$$

$$\lambda_{\text{glass}} = \frac{5.5 \times 10^{-7}}{1.5} = 3.67 \times 10^{-7} \text{ m}$$

Final Answer

(i) $f = 5.45 \times 10^8$ MHz, $T = 1.83 \times 10^{-9} \mu\text{s}$ (ii) $\lambda_{\text{glass}} = 3.67 \times 10^{-7}$ m

Common Mistake: Watch the unit conversions carefully: $1 \text{ MHz} = 10^6 \text{ Hz}$ and $1 \mu\text{s} = 10^{-6} \text{ s}$ – a very common place to lose marks.

Question 9

Light waves of frequency $5.0 \times 10^{14} \text{ Hz}$ pass through a liquid in which the wavelength of light is found to be $4.5 \times 10^{-7} \text{ m}$. Find (i) wavelength of light in vacuum, (ii) refractive index of the liquid, (iii) velocity of light in the liquid.

Given: $f = 5.0 \times 10^{14} \text{ Hz}$, $\lambda_{\text{liquid}} = 4.5 \times 10^{-7} \text{ m}$, $c = 3.0 \times 10^8 \text{ m s}^{-1}$

Formula: $\lambda_{\text{vac}} = \frac{c}{f}$, $n = \frac{\lambda_{\text{vac}}}{\lambda_{\text{liquid}}}$, $v = f\lambda_{\text{liquid}}$

Solution:

$$\lambda_{\text{vac}} = \frac{3.0 \times 10^8}{5.0 \times 10^{14}} = 6.0 \times 10^{-7} \text{ m}$$

$$n = \frac{6.0 \times 10^{-7}}{4.5 \times 10^{-7}} = 1.33$$

$$v = f\lambda_{\text{liquid}} = 5.0 \times 10^{14} \times 4.5 \times 10^{-7} = 2.25 \times 10^8 \text{ m s}^{-1}$$

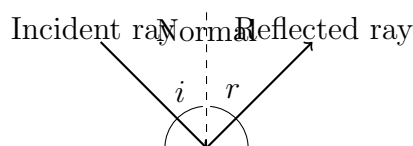
Final Answer

(i) $\lambda_{\text{vac}} = 6.0 \times 10^{-7} \text{ m}$ (ii) $n = 1.33$ (iii) $v = 2.25 \times 10^8 \text{ m s}^{-1}$

Question 10

Light of wavelength $5000 \text{ \AA}$ is incident on a plane mirror. What are the wavelength and frequency of the reflected light? For what angle of incidence is the reflected ray perpendicular to the incident ray?

★ Likely Exam Question



Incident and reflected rays at a plane mirror; reflected ray perpendicular to incident ray when $i = r = 45^\circ$

Given: $\lambda_{\text{incident}} = 5000 \text{ \AA}$, $c = 3.0 \times 10^8 \text{ m s}^{-1}$

Formula: $f = \frac{c}{\lambda}$; reflection stays in the same medium so λ is unchanged; for reflected ray $\perp$ incident ray: $i + r = 90^\circ$ with $i = r$

Solution:

Wavelength and frequency of reflected light (reflection occurs in the same medium, air):

$$\lambda_{\text{reflected}} = 5000 \text{ \AA}$$

$$f = \frac{3.0 \times 10^8}{5000 \times 10^{-10}} = 6.0 \times 10^{14} \text{ Hz}$$

For the reflected ray to be perpendicular to the incident ray, $i + r = 90^\circ$. Since $i = r$ (law of reflection),

$$2i = 90^\circ \implies i = 45^\circ$$

Final Answer

$$\lambda = 5000 \text{ \AA}, \quad f = 6 \times 10^{14} \text{ Hz}, \quad \text{angle of incidence} = 45^\circ$$

Common Mistake: Reflection never changes the wavelength or frequency of light – these change only on refraction into a new medium.

Question 11

Yellow light of wavelength 600 nm is incident on a boundary separating air and glass. Find the speed, wavelength and frequency of (i) reflected light and (ii) refracted light. The refractive index of glass for yellow light is 1.5.

Given: $\lambda_{\text{air}} = 600 \text{ nm}$, $n = 1.5$, $c = 3.0 \times 10^8 \text{ m s}^{-1}$

Formula: Reflected: stays in air, so $v = c$, λ , f unchanged. Refracted: $v = \frac{c}{n}$, $\lambda' = \frac{\lambda}{n}$, f unchanged

Solution:

(i) **Reflected light** (remains in air):

$$v = 3.0 \times 10^8 \text{ m s}^{-1}, \quad \lambda = 600 \text{ nm}, \quad f = \frac{3.0 \times 10^8}{600 \times 10^{-9}} = 5.0 \times 10^{14} \text{ Hz}$$

(ii) **Refracted light** (enters glass):

$$v' = \frac{c}{n} = \frac{3.0 \times 10^8}{1.5} = 2.0 \times 10^8 \text{ m s}^{-1}$$

$$\lambda' = \frac{\lambda}{n} = \frac{600}{1.5} = 400 \text{ nm}, \quad f' = f = 5.0 \times 10^{14} \text{ Hz}$$

Final Answer

(i) $v = 3.0 \times 10^8 \text{ m s}^{-1}$, $\lambda = 600 \text{ nm}$, $f = 5.0 \times 10^{14} \text{ Hz}$ (ii) $v' = 2.0 \times 10^8 \text{ m s}^{-1}$, $\lambda' = 400 \text{ nm}$, $f' = 5.0 \times 10^{14} \text{ Hz}$

Common Mistake: Frequency is the same for both the reflected and refracted rays – it is only the speed and wavelength that differ between the two media.

II. Superposition of Waves

Question 12

Two waves of equal frequencies have their amplitudes in the ratio 3 : 5. They are superimposed on each other. Calculate the ratio of maximum and minimum intensities of the resultant wave.

★ Likely Exam Question

Given: $a_1 : a_2 = 3 : 5$

Formula: $\frac{I_{max}}{I_{min}} = \left(\frac{a_1 + a_2}{a_1 - a_2} \right)^2$

Solution:

$$\frac{I_{max}}{I_{min}} = \left(\frac{3 + 5}{5 - 3} \right)^2 = \left(\frac{8}{2} \right)^2 = 16$$

Final Answer

$$I_{max} : I_{min} = 16 : 1$$

Common Mistake: Always take the ratio as (sum)²/(difference)² – do not square the amplitude ratio directly, as that gives the wrong intensity ratio of individual waves, not of maxima and minima.

Question 13

Two coherent light sources whose intensity ratio is 81 : 1 produce interference fringes. Calculate the ratio of intensities of maxima and minima in the fringe system.

Given: $I_1 : I_2 = 81 : 1$

Formula: $a \propto \sqrt{I}$, $\frac{I_{max}}{I_{min}} = \left(\frac{a_1 + a_2}{a_1 - a_2} \right)^2$

Solution:

$$a_1 : a_2 = \sqrt{81} : \sqrt{1} = 9 : 1$$

$$\frac{I_{max}}{I_{min}} = \left(\frac{9 + 1}{9 - 1} \right)^2 = \left(\frac{10}{8} \right)^2 = \frac{100}{64} = \frac{25}{16}$$

Final Answer

$$I_{max} : I_{min} = 25 : 16$$

Question 14

Two beams of light of the same wavelength and having intensities I and $4I$ interfere to produce a fringe pattern on a screen. The path difference between the beams is $\lambda/2$ at point P and λ at point Q on the screen. Find the ratio of the resultant intensities at P and Q .

Given: $I_1 = I$, $I_2 = 4I$, $\Delta x_P = \lambda/2$, $\Delta x_Q = \lambda$

Formula: $\phi = \frac{2\pi}{\lambda} \Delta x$, $I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$

Solution:

At P : $\Delta x = \lambda/2 \implies \phi_P = \pi$, so $\cos \phi_P = -1$.

$$I_P = I + 4I + 2\sqrt{I \times 4I}(-1) = 5I - 4I = I$$

At Q : $\Delta x = \lambda \implies \phi_Q = 2\pi$, so $\cos \phi_Q = 1$.

$$I_Q = I + 4I + 2\sqrt{I \times 4I}(1) = 5I + 4I = 9I$$

$$\frac{I_P}{I_Q} = \frac{I}{9I} = \frac{1}{9}$$

Final Answer

$$\frac{I_P}{I_Q} = \frac{1}{9}$$

Common Mistake: A path difference of λ still gives constructive interference (not $\lambda/2$ giving maxima) – match path difference to phase using $\phi = 2\pi\Delta x/\lambda$ before applying $\cos \phi$.

III. Young's Double Slit Experiment (YDSE)

Question 15

The initial phase difference between two interfering waves is 54° . When these waves reach a point on a screen, a path difference of 1.5λ is introduced between them. What is the total phase difference at that point?

Given: Initial phase = 54° , additional path difference = 1.5λ

Formula: $\phi = \left(54^\circ \times \frac{\pi}{180^\circ}\right) + \left(1.5\lambda \times \frac{2\pi}{\lambda}\right)$

Solution:

Phase from initial condition:

$$54^\circ = 54 \times \frac{\pi}{180} = 0.3\pi \text{ rad}$$

Phase from the path difference 1.5λ :

$$\phi_{\text{path}} = \frac{2\pi}{\lambda} \times 1.5\lambda = 3\pi \text{ rad}$$

Total phase difference:

$$\phi = 0.3\pi + 3\pi = 3.3\pi \text{ rad}$$

Final Answer

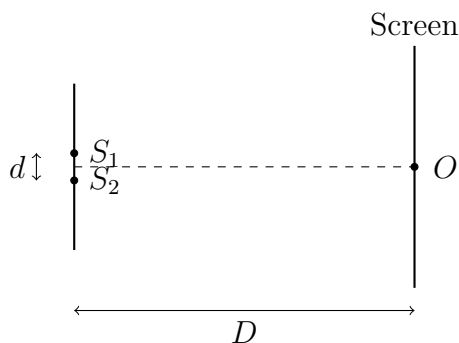
$$\phi = 3.3\pi \text{ rad}$$

Common Mistake: Convert the initial phase from degrees to radians (as a multiple of π) before adding it to the path-difference term, otherwise the two parts cannot be combined.

Question 16

A double-slit of separation 0.5 mm is illuminated by blue light of wavelength 480 nm. At what distance should a screen be placed from the double-slit to obtain interference fringes that are 1.0 mm apart? What would be the fringe-width if this distance is doubled?

★ Likely Exam Question



Standard YDSE geometry: two slits S_1, S_2 separated by d , screen at distance D

Given: $d = 0.5 \text{ mm} = 5 \times 10^{-4} \text{ m}$, $\lambda = 480 \text{ nm} = 4.8 \times 10^{-7} \text{ m}$, $\beta = 1.0 \text{ mm} = 1 \times 10^{-3} \text{ m}$

Formula: $\beta = \frac{\lambda D}{d} \implies D = \frac{\beta d}{\lambda}$

Solution:

$$D = \frac{(1 \times 10^{-3})(5 \times 10^{-4})}{4.8 \times 10^{-7}} = \frac{5 \times 10^{-7}}{4.8 \times 10^{-7}} = 1.04 \text{ m}$$

If D is doubled, since $\beta \propto D$:

$$\beta' = 2\beta = 2.0 \text{ mm}$$

Final Answer

$D = 104 \text{ cm}$; new fringe width = 2.0 mm

Question 17

Two slits are 1.0 mm apart and placed at a distance of 1.0 m from a screen. Find the fringe-width when light of wavelength 500 nm is used. What would be the fringe-width if the separation between the slits is halved and the distance of the screen from the slits is doubled?

Given: $d = 1.0 \text{ mm} = 1 \times 10^{-3} \text{ m}$, $D = 1.0 \text{ m}$, $\lambda = 500 \text{ nm} = 5 \times 10^{-7} \text{ m}$

Formula: $\beta = \frac{\lambda D}{d}$

Solution:

$$\beta = \frac{5 \times 10^{-7} \times 1.0}{1 \times 10^{-3}} = 5 \times 10^{-4} \text{ m} = 0.5 \text{ mm}$$

New case: $d' = d/2$, $D' = 2D$:

$$\beta' = \frac{\lambda D'}{d'} = \frac{\lambda(2D)}{(d/2)} = 4 \left(\frac{\lambda D}{d} \right) = 4\beta = 4 \times 0.5 = 2.0 \text{ mm}$$

Final Answer

$\beta = 0.5 \text{ mm}$; new fringe width = 2.0 mm

Common Mistake: Both halving d and doubling D increase β – their effects multiply (factor of $2 \times 2 = 4$), not add.

Question 18

In Young's experiment, what will be the phase difference and the path difference between the light waves reaching (i) the third bright fringe and (ii) the third dark fringe from the central fringe? ($\lambda = 6000 \text{ \AA}$)

Given: $\lambda = 6000 \text{ \AA}$

Formula: Bright fringe: $\Delta x = n\lambda$; Dark fringe: $\Delta x = (2n - 1)\frac{\lambda}{2}$; $\phi = \frac{2\pi}{\lambda}\Delta x$

Solution:

(i) Third bright fringe ($n = 3$):

$$\Delta x = 3\lambda = 3 \times 6000 = 18000 \text{ \AA}$$

$$\phi = \frac{2\pi}{\lambda} \times 3\lambda = 6\pi \text{ rad}$$

(ii) Third dark fringe ($n = 3$):

$$\Delta x = \frac{(2 \times 3 - 1)\lambda}{2} = \frac{5\lambda}{2} = 2.5 \times 6000 = 15000 \text{ \AA}$$

$$\phi = \frac{2\pi}{\lambda} \times 2.5\lambda = 5\pi \text{ rad}$$

Final Answer

(i) $\phi = 6\pi \text{ rad}$, $\Delta x = 18000 \text{ \AA}$ (ii) $\phi = 5\pi \text{ rad}$, $\Delta x = 15000 \text{ \AA}$

Common Mistake: The n th dark fringe corresponds to path difference $(2n - 1)\lambda/2$, not $n\lambda/2$ – a very common slip.

Question 19

In Young's double-slit experiment, the slits are 0.2 mm apart and the screen is 1.5 m away. The distance between the central bright fringe and the fourth dark fringe is 1.8 cm. Find the wavelength of light used and the fringe width.

★ Likely Exam Question

Given: $d = 0.2 \text{ mm} = 2 \times 10^{-4} \text{ m}$, $D = 1.5 \text{ m}$, $x_4(\text{dark}) = 1.8 \text{ cm} = 1.8 \times 10^{-2} \text{ m}$

Formula: $x_n(\text{dark}) = \frac{(2n-1)\lambda D}{2d}, \quad \beta = \frac{\lambda D}{d}$

Solution:

For the 4th dark fringe, $n = 4$:

$$x_4 = \frac{(2 \times 4 - 1)\lambda D}{2d} = \frac{7\lambda D}{2d}$$

$$1.8 \times 10^{-2} = \frac{7 \times \lambda \times 1.5}{2 \times 2 \times 10^{-4}}$$

$$\lambda = \frac{1.8 \times 10^{-2} \times 2 \times 2 \times 10^{-4}}{7 \times 1.5} = \frac{7.2 \times 10^{-6}}{10.5} = 6.857 \times 10^{-7} \text{ m} = 6857 \text{ \AA}$$

$$\beta = \frac{\lambda D}{d} = \frac{6.857 \times 10^{-7} \times 1.5}{2 \times 10^{-4}} = 5.14 \times 10^{-3} \text{ m} = 5.14 \text{ mm}$$

Final Answer

$$\lambda = 6857 \text{ \AA}, \quad \beta = 5.14 \text{ mm}$$

Question 20

The light used in Young's double-slit experiment has a frequency of $6.0 \times 10^{14} \text{ s}^{-1}$. The distance between the centres of adjacent bright fringes is 0.75 mm. If the screen is 1.5 m away, what is the distance between the slits?

Given: $f = 6.0 \times 10^{14} \text{ Hz}, \quad \beta = 0.75 \text{ mm} = 7.5 \times 10^{-4} \text{ m}, \quad D = 1.5 \text{ m}$

Formula: $\lambda = \frac{c}{f}, \quad d = \frac{\lambda D}{\beta}$

Solution:

$$\lambda = \frac{3.0 \times 10^8}{6.0 \times 10^{14}} = 5.0 \times 10^{-7} \text{ m}$$

$$d = \frac{\lambda D}{\beta} = \frac{5.0 \times 10^{-7} \times 1.5}{7.5 \times 10^{-4}} = 1.0 \times 10^{-3} \text{ m} = 1.0 \text{ mm}$$

Final Answer

$$d = 1.0 \text{ mm}$$

Question 21

In Young's double-slit experiment, the slits are separated by 0.24 mm. The screen is 1.2 m away from the slits. The fringe-width is 0.3 cm. Find the wavelength of light used.

Given: $d = 0.24 \text{ mm} = 2.4 \times 10^{-4} \text{ m}, \quad D = 1.2 \text{ m}, \quad \beta = 0.3 \text{ cm} = 3 \times 10^{-3} \text{ m}$

Formula: $\lambda = \frac{\beta d}{D}$

Solution:

$$\lambda = \frac{3 \times 10^{-3} \times 2.4 \times 10^{-4}}{1.2} = \frac{7.2 \times 10^{-7}}{1.2} = 6.0 \times 10^{-7} \text{ m} = 6000 \text{ \AA}$$

Final Answer

$$\lambda = 6000 \text{ \AA}$$

Question 22

In Young's experiment, the wavelength of light used is $6000 \text{ \AA}$, the distance between two slits is 0.05 cm and the distance of the screen from the slits is 1 m . Find the distance between the first dark fringe and first bright fringe.

Given: $\lambda = 6000 \text{ \AA} = 6 \times 10^{-7} \text{ m}$, $d = 0.05 \text{ cm} = 5 \times 10^{-4} \text{ m}$, $D = 1 \text{ m}$

Formula: $\beta = \frac{\lambda D}{d}$; distance between adjacent dark and bright fringe $= \frac{\beta}{2}$

Solution:

$$\beta = \frac{6 \times 10^{-7} \times 1}{5 \times 10^{-4}} = 1.2 \times 10^{-3} \text{ m} = 1.2 \text{ mm}$$

Since a dark fringe always lies exactly midway between two consecutive bright fringes,

$$\text{distance} = \frac{\beta}{2} = \frac{1.2}{2} = 0.6 \text{ mm} = 0.06 \text{ cm}$$

Final Answer

$$\text{Distance} = 0.06 \text{ cm}$$

Common Mistake: The separation between a bright fringe and the *nearest* dark fringe is always $\beta/2$, not β .

Question 23

In Young's double-slit experiment, the separation between the slits is 0.5 mm and the distance between the slits and the screen is 1.5 m . If the fringe width is 1.8 mm , find the path difference between the interfering waves at the position of the first-order maximum.

Given: $d = 0.5 \text{ mm} = 5 \times 10^{-4} \text{ m}$, $D = 1.5 \text{ m}$, $\beta = 1.8 \text{ mm} = 1.8 \times 10^{-3} \text{ m}$

Formula: $\lambda = \frac{\beta d}{D}$; path difference at n th order maximum $= n\lambda$

Solution:

$$\lambda = \frac{1.8 \times 10^{-3} \times 5 \times 10^{-4}}{1.5} = \frac{9 \times 10^{-7}}{1.5} = 6.0 \times 10^{-7} \text{ m}$$

At the first-order maximum ($n = 1$), path difference = $n\lambda = \lambda$:

$$\Delta x = 6.0 \times 10^{-7} \text{ m} = 6000 \text{ \AA}$$

Final Answer

$$\Delta x = 6 \times 10^{-7} \text{ m}$$

IV. Angular Fringe Width

Question 24

A double-slit is illuminated by light of wavelength $6000 \text{ \AA}$. The slits are 0.1 cm apart and the screen is placed 1 m away. Find the angular position of the 10th bright fringe in radian and the angular separation between two adjacent minima.

★ Likely Exam Question

Given: $\lambda = 6000 \text{ \AA} = 6 \times 10^{-7} \text{ m}$, $d = 0.1 \text{ cm} = 1 \times 10^{-3} \text{ m}$

Formula: $\theta_n = \frac{n\lambda}{d}$, $\Delta\theta = \frac{\lambda}{d}$

Solution:

$$\theta_{10} = \frac{10 \times 6 \times 10^{-7}}{1 \times 10^{-3}} = 6 \times 10^{-3} \text{ rad}$$

$$\Delta\theta = \frac{\lambda}{d} = \frac{6 \times 10^{-7}}{1 \times 10^{-3}} = 6 \times 10^{-4} \text{ rad} = 0.6 \text{ mrad}$$

Final Answer

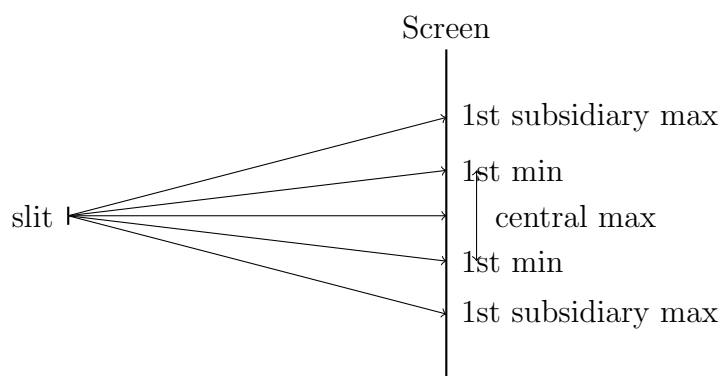
$\theta_{10} = 6 \times 10^{-3} \text{ rad}$; angular separation between adjacent minima $= 0.6 \times 10^{-3} \text{ rad}$

Common Mistake: Notice that the screen distance D is not needed here – the angular position depends only on λ and d , since $\theta = x/D = n\lambda D/(dD) = n\lambda/d$.

V. Diffraction at a Single Slit

Question 25

A parallel beam of light of wavelength 600 nm falls normally on a narrow slit of width 0.3 mm. Calculate the angular separation between (a) the first minimum and the central maximum, and (b) the first subsidiary maxima on the two sides of the central maximum.



Single-slit diffraction: central maximum flanked by first minima and first subsidiary maxima

Given: $\lambda = 600 \text{ nm} = 6 \times 10^{-7} \text{ m}$, $a = 0.3 \text{ mm} = 3 \times 10^{-4} \text{ m}$

Formula: $\theta_{\min} = \frac{\lambda}{a}$; first subsidiary maximum: $\theta = \frac{3\lambda}{2a}$

Solution:

(a) Angular separation between first minimum and central maximum:

$$\theta_1 = \frac{\lambda}{a} = \frac{6 \times 10^{-7}}{3 \times 10^{-4}} = 2 \times 10^{-3} \text{ rad}$$

(b) Angular position of the first subsidiary maximum on one side:

$$\theta = \frac{3\lambda}{2a} = \frac{3 \times 6 \times 10^{-7}}{2 \times 3 \times 10^{-4}} = 3 \times 10^{-3} \text{ rad}$$

Angular separation between the first subsidiary maxima on the *two* sides:

$$2\theta = 2 \times 3 \times 10^{-3} = 6 \times 10^{-3} \text{ rad}$$

Final Answer

(a) $2 \times 10^{-3} \text{ rad}$ (b) $6 \times 10^{-3} \text{ rad}$

Common Mistake: Part (b) asks for the separation between the subsidiary maxima on *both* sides – remember to double the single-side angle.

Question 26

A Fraunhofer diffraction pattern due to a single slit of width 0.2 mm is obtained on a screen placed at a distance of 2 m from the slit. The first minima lie 5 mm on either side of the central maximum on the screen. Find the wavelength of light.

★ Likely Exam Question

Given: $a = 0.2 \text{ mm} = 2 \times 10^{-4} \text{ m}$, $D = 2 \text{ m}$, $x = 5 \text{ mm} = 5 \times 10^{-3} \text{ m}$

Formula: $x = \frac{\lambda D}{a}$

Solution:

$$\lambda = \frac{xa}{D} = \frac{5 \times 10^{-3} \times 2 \times 10^{-4}}{2} = 5 \times 10^{-7} \text{ m} = 5000 \text{ \AA}$$

Final Answer

$$\lambda = 5000 \text{ \AA}$$

Question 27

A parallel beam of light of wavelength $6 \times 10^{-5} \text{ cm}$ falls normally on a straight slit of width 0.2 mm. Find the total angular width of the central diffraction maximum and also its linear width as observed on a screen placed 2 m away.

Given: $\lambda = 6 \times 10^{-5} \text{ cm} = 6 \times 10^{-7} \text{ m}$, $a = 0.2 \text{ mm} = 2 \times 10^{-4} \text{ m}$, $D = 2 \text{ m}$

Formula: Angular width $= \frac{2\lambda}{a}$; linear width $= \frac{2\lambda D}{a}$

Solution:

$$\text{Angular width} = \frac{2 \times 6 \times 10^{-7}}{2 \times 10^{-4}} = 6 \times 10^{-3} \text{ rad}$$

$$\text{Linear width} = \frac{2\lambda D}{a} = 6 \times 10^{-3} \times 2 = 1.2 \times 10^{-2} \text{ m} = 1.2 \text{ cm}$$

Final Answer

$$\text{Angular width} = 6 \times 10^{-3} \text{ rad}; \text{ linear width} = 1.2 \text{ cm}$$

Question 28

In a single-slit diffraction experiment, the first minimum for $\lambda_1 = 600$ nm coincides with the first maxima for wavelength λ_2 . Calculate λ_2 .

Given: $\lambda_1 = 600$ nm; first minimum of λ_1 coincides with first (subsidiary) maximum of λ_2

Formula: First minimum: $a \sin \theta = \lambda_1$; first subsidiary maximum: $a \sin \theta = \frac{3\lambda_2}{2}$

Solution:

Since both occur at the same angle θ ,

$$\lambda_1 = \frac{3\lambda_2}{2} \implies \lambda_2 = \frac{2\lambda_1}{3}$$

$$\lambda_2 = \frac{2 \times 600}{3} = 400 \text{ nm}$$

Final Answer

$$\lambda_2 = 400 \text{ nm}$$

Common Mistake: Use the *approximate* secondary-maximum condition $a \sin \theta = (2n + 1)\lambda/2$ with $n = 1$, i.e. $3\lambda/2$, not λ – confusing this with the minima condition is a common error.

VI. Miscellaneous

Question 29

In Young's double-slit experiment, the distance between the centres of the adjacent fringes is 0.10 mm. If the distance of the screen from the slits is doubled, the distance between the slits is halved, and the wavelength of light is changed from 6.4×10^{-7} m to 4.0×10^{-7} m, what is the new distance between the fringes?

Given: $\beta_1 = 0.10$ mm, $\lambda_1 = 6.4 \times 10^{-7}$ m, $\lambda_2 = 4.0 \times 10^{-7}$ m, $D_2 = 2D_1$, $d_2 = d_1/2$

Formula: $\beta = \frac{\lambda D}{d}$

Solution:

$$\frac{\beta_2}{\beta_1} = \frac{\lambda_2}{\lambda_1} \times \frac{D_2}{D_1} \times \frac{d_1}{d_2} = \frac{\lambda_2}{\lambda_1} \times 2 \times 2 = \frac{4\lambda_2}{\lambda_1}$$

$$\frac{\beta_2}{\beta_1} = \frac{4 \times 4.0 \times 10^{-7}}{6.4 \times 10^{-7}} = 2.5$$

$$\beta_2 = 2.5 \times 0.10 = 0.25 \text{ mm}$$

Final Answer

New fringe separation = 0.25 mm

Common Mistake: Track all three changes (D , d , λ) together as ratios multiplying β_1 – do not apply them one at a time and round in between, as this accumulates error.

Question 30

In Young's double-slit experiment, the slits are 1 mm apart and the screen is placed 1 m away from the slits. When the slits are illuminated by monochromatic light, the width of 10 fringes on the screen is measured as 3.93 mm. Find the wavelength of light used. What will be the width of 10 fringes when the screen is placed 1.5 m away from the slits and the same light is used?

Given: $d = 1$ mm = 1×10^{-3} m, $D = 1$ m, $10\beta = 3.93$ mm

Formula: $\beta = \frac{\lambda D}{d}$

Solution:

$$\beta = \frac{3.93}{10} = 0.393 \text{ mm} = 3.93 \times 10^{-4} \text{ m}$$

$$\lambda = \frac{\beta d}{D} = \frac{3.93 \times 10^{-4} \times 1 \times 10^{-3}}{1} = 3.93 \times 10^{-7} \text{ m} = 3930 \text{ \AA}$$

Since $\beta \propto D$ (with d, λ unchanged), when $D' = 1.5 \text{ m}$:

$$10\beta' = 10\beta \times \frac{D'}{D} = 3.93 \times 1.5 = 5.895 \text{ mm}$$

Final Answer

$\lambda = 3930 \text{ \AA}$; new width of 10 fringes = 5.895 mm

Common Mistake: You do not need to recompute λ for the second part – since β is directly proportional to D , just scale the given fringe width by the ratio of the two distances.

Question 31

In an experiment with sodium light ($5890 \text{ \AA}$), an interference pattern is obtained in which 20 equally spaced fringes occupy 2.30 cm on the screen. On replacing sodium light with another monochromatic source, 30 fringes occupy 2.80 cm on the screen. Find the wavelength of the light from the second source.

Given: $\lambda_1 = 5890 \text{ \AA}$; $20\beta_1 = 2.30 \text{ cm}$; $30\beta_2 = 2.80 \text{ cm}$

Formula: $\beta = \frac{\lambda D}{d} \implies \beta \propto \lambda$ (for fixed D, d)

Solution:

$$\beta_1 = \frac{2.30}{20} = 0.115 \text{ cm}, \quad \beta_2 = \frac{2.80}{30} = 0.0933 \text{ cm}$$

Since $\beta \propto \lambda$ for the same apparatus,

$$\lambda_2 = \lambda_1 \times \frac{\beta_2}{\beta_1} = 5890 \times \frac{0.0933}{0.115} = 5890 \times 0.8116$$

$$\lambda_2 \approx 4780 \text{ \AA}$$

Final Answer

$\lambda_2 \approx 4780 \text{ \AA}$

Common Mistake: Compute the fringe width for each case separately (total width $\div$ number of fringes) before taking the ratio – do not divide the two total widths directly, since the fringe counts differ.

Question 32

A double-slit is illuminated by sodium light ($\lambda = 5893 \text{ \AA}$). In the double-slit, the distance between the two slits is 0.050 cm . The interference pattern is obtained on a screen placed at a distance of 1.00 m . Calculate the distance between the fifth bright fringe and the first dark fringe on the same side of the central fringe.

Given: $\lambda = 5893 \text{ \AA} = 5.893 \times 10^{-7} \text{ m}$, $d = 0.050 \text{ cm} = 5 \times 10^{-4} \text{ m}$, $D = 1.00 \text{ m}$

Formula: $x_n(\text{bright}) = \frac{n\lambda D}{d}$; $x_n(\text{dark}) = \frac{(2n-1)\lambda D}{2d}$

Solution:

Fringe width:

$$\beta = \frac{\lambda D}{d} = \frac{5.893 \times 10^{-7} \times 1.00}{5 \times 10^{-4}} = 1.1786 \times 10^{-3} \text{ m}$$

Position of the 5th bright fringe ($n = 5$): $x_5 = 5\beta$. Position of the 1st dark fringe ($n = 1$):

$$x_1(\text{dark}) = \frac{\beta}{2}.$$

$$\Delta x = x_5 - x_1(\text{dark}) = 5\beta - \frac{\beta}{2} = 4.5\beta$$

$$\Delta x = 4.5 \times 1.1786 \times 10^{-3} = 5.30 \times 10^{-3} \text{ m} \approx 0.53 \text{ cm}$$

Final Answer

Distance $\approx 0.53 \text{ cm}$

Common Mistake: The first dark fringe lies at $\beta/2$ from the centre (not at β) – always use $(2n-1)\beta/2$ for dark fringes and $n\beta$ for bright fringes, and be careful not to mix the two.

Question 33

In Young's double-slit experiment, the separation between the slits is 0.15 mm and the fringes are obtained on a screen distant 0.75 m from the slits. If the third dark fringe is at a distance of 5.5 mm from the central bright fringe, find the wavelength of light used. What will be the distance of the same dark fringe from the centre if the entire apparatus is immersed in water? (Refractive index of water = $4/3$)

Given: $d = 0.15 \text{ mm} = 1.5 \times 10^{-4} \text{ m}$, $D = 0.75 \text{ m}$, $x_3(\text{dark}) = 5.5 \text{ mm}$, $n_w = 4/3$

Formula: $x_n(\text{dark}) = \frac{(2n-1)\lambda D}{2d}$; $\lambda_{\text{water}} = \frac{\lambda_{\text{air}}}{n_w}$

Solution:

For the 3rd dark fringe, $n = 3$:

$$x_3 = \frac{(2 \times 3 - 1)\lambda D}{2d} = \frac{5\lambda D}{2d}$$

$$5.5 \times 10^{-3} = \frac{5 \times \lambda \times 0.75}{2 \times 1.5 \times 10^{-4}}$$

$$\lambda = \frac{5.5 \times 10^{-3} \times 2 \times 1.5 \times 10^{-4}}{5 \times 0.75} = \frac{1.65 \times 10^{-6}}{3.75} = 4.4 \times 10^{-7} \text{ m} = 4400 \text{ \AA}$$

When immersed in water, D and d are unchanged but λ decreases:

$$\lambda_{\text{water}} = \frac{4400}{4/3} = 3300 \text{ \AA}$$

Since $x_3 \propto \lambda$,

$$x'_3 = x_3 \times \frac{\lambda_{\text{water}}}{\lambda_{\text{air}}} = 5.5 \times \frac{3300}{4400} = 5.5 \times 0.75 = 4.125 \text{ mm}$$

Final Answer

$\lambda = 4400 \text{ \AA}$; new distance of the same dark fringe = 4.125 mm

Common Mistake: On immersion in a liquid, D and d do not change – only λ shrinks by a factor of n_w , so all fringe positions shrink by the same factor.

Question 34

A parallel light beam of wavelength $6000 \text{ \AA}$ passes through a slit 0.2 mm wide and forms a diffraction pattern on a screen 1.0 m away from the slit. Find the width of the central maximum on the screen. What would be the width of this maximum if the apparatus were immersed in water of refractive index $4/3$?

Given: $\lambda = 6000 \text{ \AA} = 6 \times 10^{-7} \text{ m}$, $a = 0.2 \text{ mm} = 2 \times 10^{-4} \text{ m}$, $D = 1.0 \text{ m}$, $n_w = 4/3$

Formula: Width of central maximum = $\frac{2\lambda D}{a}$

Solution:

$$w = \frac{2 \times 6 \times 10^{-7} \times 1.0}{2 \times 10^{-4}} = 6 \times 10^{-3} \text{ m} = 6.0 \text{ mm}$$

In water, $\lambda_{\text{water}} = \frac{\lambda}{n_w} = \frac{6000}{4/3} = 4500 \text{ \AA} = 4.5 \times 10^{-7} \text{ m}$:

$$w' = \frac{2 \times 4.5 \times 10^{-7} \times 1.0}{2 \times 10^{-4}} = 4.5 \times 10^{-3} \text{ m} = 4.5 \text{ mm}$$

Final Answer

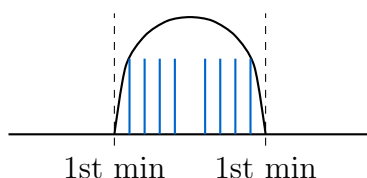
$w = 6.0 \text{ mm}$ in air; $w' = 4.5 \text{ mm}$ in water

Question 35

In a double-slit experiment, the two slits are 1 mm apart and the screen is placed 1 m away. A monochromatic light of wavelength 500 nm is used. What will be the width of each slit for obtaining ten maxima of the double-slit pattern within the central maximum of the single-slit diffraction pattern?

★ Likely Exam Question

central diffraction envelope



Double-slit fringes (fine lines) lying inside the broad single-slit central diffraction envelope

Given: $d = 1 \text{ mm} = 1 \times 10^{-3} \text{ m}$, $D = 1 \text{ m}$, $\lambda = 500 \text{ nm} = 5 \times 10^{-7} \text{ m}$, $N = 10$ maxima

Formula: Width of central max (single slit) $= \frac{2\lambda D}{e}$; fringe width (double slit) $\beta = \frac{\lambda D}{d}$; $N = \frac{2\lambda D/e}{\lambda D/d} = \frac{2d}{e}$

Solution:

The number of double-slit maxima that fit within the single-slit central maximum is

$$N = \frac{2d}{e}$$

Setting $N = 10$:

$$10 = \frac{2d}{e} \Rightarrow e = \frac{2d}{10} = \frac{d}{5}$$

$$e = \frac{1 \text{ mm}}{5} = 0.2 \text{ mm}$$

Final Answer

Slit width $e = 0.2 \text{ mm}$

Common Mistake: This problem does not actually require λ or D in the final step – both cancel out, leaving a simple ratio between d and e . Plugging in numbers unnecessarily often leads to arithmetic errors.

Question 36

A narrow slit of width 0.3 mm is illuminated normally by a parallel beam of light of wavelength 6×10^{-7} m. The Fraunhofer diffraction pattern is obtained on a screen placed in the focal plane of a convex lens of focal length 25 cm. The lens is placed quite close to the slit. Find the width of the central maximum on the screen.

Given: $a = 0.3 \text{ mm} = 3 \times 10^{-4} \text{ m}$, $\lambda = 6 \times 10^{-7} \text{ m}$, $f = 25 \text{ cm} = 0.25 \text{ m}$

Formula: Width of central maximum $= \frac{2f\lambda}{a}$

Solution:

Since the lens is close to the slit and the screen is at the focal plane, D is effectively replaced by f :

$$w = \frac{2 \times 0.25 \times 6 \times 10^{-7}}{3 \times 10^{-4}} = \frac{3 \times 10^{-7}}{3 \times 10^{-4}} = 1 \times 10^{-3} \text{ m} = 1.0 \text{ mm}$$

Final Answer

$$w = 1.0 \text{ mm}$$

Common Mistake: When a converging lens focuses the diffraction pattern, use its focal length f in place of the slit-to-screen distance D – do not add a separate screen distance.